KING MONGKUT'S UNIVERSITY OF TECHNOLOGY THONBURI | DEPARTMENT OF COMPUTER ENGINEERING

CPE 334 Introduction to Software Engineering in the Age of AI Agents (Semester 1/2026)

Pollipop

University Review, Feedback, and Survey System

Project Deliverable 1: Engineering Baseline and Working Sprint 1 (Full Engineering Report & Peer Review Evidence)

Course & Term:	**CPE 334** (Semester 1/2026) | Due: 8 November 2026, 23:59 (80 Points / 8% Grade)
Assigned Theme:	**Theme 10: Review, Feedback, and Survey System**
GitHub Repository:	`https://github.com/phatthidawadi/pollipop` (Public per requirement R-12)
Release Branch & Commit:	Branch `sprint-1-staging` | Commit `6038d20` | Release Tag `v1.0.0-sprint1`
Student 1 (Author / Collaborator):	ภัทร์ธิดาวดี อุ่นคำ (Phatthidawadi Aunkham) — รหัสนักศึกษา: **67070505225** (GitHub: `@phatthidawadi`)
Student 2 (Author / Collaborator):	ปทิตญา แก้ววิเชียร (Patitaya Kaewwichian) — รหัสนักศึกษา: **67070505220** (GitHub: `@lmaybelgracel`)
Target Score:	**80 / 80 Points (100%)** with Complete Peer Review Evidence across 11 PRs

Master Evidence Index & Short Navigation Table (Protocol Section 4.3)

In full adherence to CPE 334 Project Deliverable 1 Protocol Section 4.3, this master navigation table links every core part of the engineering delivery to its stable requirement IDs, artifact files, and verifiable execution evidence:

Part	Protocol Heading & Scope	Core Stable IDs	Artifact Source & Verifiable Evidence
Part A	Software Requirements Specification (SRS)	FR-01..16, BR-01..13, NFR-01..05	`docs/srs.md` (Domain context, actors, scope, decisions)
Part B	Feature Inventory & Traceability Matrix	S-01, F-01..F-09	`docs/features.md`, `docs/traceability.md` (Full requirements mapping)
Part C	System-Level Software Design Specification	Three-Tier Architecture, ERD	`docs/system-sds.md` (Stack, ERD, Security, Role Matrix, GCP Plan)
Part D	Three-Sprint Plan & Pair Workflow	Sprints 1, 2, 3	`docs/sprints/plan.md` (Vertical slice rationale, 10 balanced issues)
Part E	Sprint 1 Engineering Contracts & DoD	AC-01..38, TEST-01..40	`docs/sprints/sprint-1/contracts.md` (Feature design, STS, DoD)
Part F	Working Sprint 1 Verification & Peer Reviews	46 Tests (100% Pass) + 11 PRs	`docs/sprints/sprint-1/verification.md` (9 Screenshots, 11 PR reviews)

Part A: Software Requirements Specification (SRS)

> **Suggested Artifact:** `docs/srs.md` | **Weight:** 10 Points | Demonstrates: Stakeholders, Context, Actors, Product Scope, Numbered FR/BR/NFR, Assumptions & Decisions.

1. Stakeholder Problem & System Context

1.1 Stakeholder Problem Statement

Higher education institutions, academic faculties, student affair departments, course lecturers, and student clubs (Survey Owners) routinely conduct evaluations regarding course instruction, library learning environments, dining services, campus Wi-Fi infrastructure, and student activities. However, relying on generic paper forms or consumer-grade online survey tools introduces significant software engineering and data integrity failures:

- **Uncontrolled Duplicate Submissions & Identity Leakage:** Standard forms either allow unrestricted, untraceable repeat submissions or tightly bind student accounts to their answers, eliminating genuine anonymity and causing social desirability bias in sensitive campus feedback.
- **Absence of Role-Based & Institutional Access Boundaries:** Generic tools lack fine-grained role segregation. An instructor can inadvertently view, modify, or delete another instructor's course evaluations, violating privacy and departmental isolation.
- **Data Corruption Caused by Post-Publish Mutations:** Allowing question options or ratings to be modified while responses are being actively submitted breaks historical validity and produces corrupted aggregate summaries.

1.2 Chosen Domain Context: Pollipop

Pollipop is a modern, cloud-ready three-tier web application specifically engineered for university course reviews and campus evaluations. The system manages the full survey lifecycle: structured survey creation (Single Choice and Rating Scale), lifecycle state transitions (`DRAFT` → `PUBLISHED` → `CLOSED`), authenticated response submission within atomic database transactions, strict duplicate prevention, and real-time server-side aggregate calculation.

The system is packaged with multi-stage Docker containers, making it ready for horizontal autoscaling on Google Cloud Platform (Cloud Run and Cloud SQL) per CPE 334 Lab 5 requirements.

2. Actors and Role-Based Responsibilities

Actor / Role	Role Code	Allowed Operations	Explicitly Forbidden Operations
Survey Owner (Instructor / Staff)	`OWNER`	- Create new surveys in `DRAFT` status with Single Choice (2–10 options) and Rating (1–5) questions. - Inspect, update (while in `DRAFT` status only), and transition survey state to `PUBLISHED` or `CLOSED`. - Inspect real-time server-calculated aggregate analytics of surveys they own.	- **Forbidden:** Accessing, modifying, publishing, or inspecting summaries of surveys owned by other owners (`403 Forbidden`). - **Forbidden:** Submitting responses to any survey as a respondent (`403 Forbidden`).

Actor / Role	Role Code	Allowed Operations	Explicitly Forbidden Operations
Survey Respondent (Student / University Member)	`RESPONDENT`	- View open (`PUBLISHED`) surveys with personal completion status (`answered: true / false`). - Submit a validated response exactly once per survey. - Review own submitted answers in `IDENTIFIED` mode via `GET /api/responses/:id` .	- **Forbidden:** Creating, modifying, publishing, or closing surveys (`403 Forbidden`). - **Forbidden:** Viewing `DRAFT` surveys in listing or detail queries (`404 Not Found`). - **Forbidden:** Viewing aggregate summary analytics (`403 Forbidden`). - **Forbidden:** Viewing other respondents' submissions (`403 Forbidden`).

3. Numbered Functional Requirements (FR)

FR ID	Observable Requirement Behavior	Sprint
FR-01	**User Login & Session Creation:** Authenticate via email/password, verify bcrypt hash, set HTTP-only `pollipop_sid` cookie.	Sprint 1
FR-02	**Current User Profile & Logout:** Retrieve profile (`GET /api/auth/me`) and destroy session (`POST /api/auth/logout`) with cache invalidation.	Sprint 1
FR-03	**Health & Readiness Probes:** `GET /api/health` (liveness) and `GET /api/ready` (readiness with database `SELECT 1`).	Sprint 1
FR-04	**Create Draft Survey:** `OWNER` creates survey with title, description, mode (`IDENTIFIED`), and questions in `DRAFT` status.	Sprint 1
FR-05	**Sprint 1 Question Types:** Accept `SINGLE_CHOICE` (2–10 options) and `RATING` (1–5) questions with `required` flag.	Sprint 1
FR-06	**Publish Survey:** `OWNER` transitions survey from `DRAFT` to `PUBLISHED` (making it active for respondents).	Sprint 1
FR-07	**List Surveys by Role:** `OWNER` sees own surveys; `RESPONDENT` sees `PUBLISHED` surveys with `answered` and `responseId` .	Sprint 1
FR-08	**Submit Validated Response:** `RESPONDENT` submits answers inside an atomic database transaction.	Sprint 1
FR-09	**View Own Submitted Response:** In `IDENTIFIED` mode, `RESPONDENT` inspects submitted answers via `GET /api/responses/:id` .	Sprint 1
FR-10	**View Survey Aggregate Summary:** `OWNER` retrieves server-calculated statistics (counts, percentages, mean rating).	Sprint 1
FR-11	**Edit Draft Survey:** `OWNER` updates title, description, or questions while in `DRAFT` status.	Sprint 2
FR-12	**Close Survey & Expiration:** Manual close (`POST /api/surveys/:id/close`) or scheduled closing (`closesAt`).	Sprint 1 / 2
FR-13	**Anonymous Survey Mode:** Decouple participation tracking from answer storage (`respondentId = null`).	Sprint 2
FR-14	**Department Eligibility:** Restrict survey answering to students matching `targetDepartment` .	Sprint 2
FR-15	**Extended Question Types:** Multi-select (`MULTIPLE_CHOICE`) and open feedback (`SHORT_TEXT`).	Sprint 2

FR ID	Observable Requirement Behavior	Sprint
FR-16	**Export Survey Results:** Download summary analytics and anonymized responses as CSV files.	Sprint 3

4. Numbered Business Rules (BR)

BR ID	Domain Rule Description	Enforced Layer
BR-01	**Single Submission per Respondent:** Duplicate attempts return `409 Conflict (ALREADY_SUBMITTED)` backed by DB unique constraint `(surveyId, userId)` on `SurveyParticipation`.	Application & DB
BR-02	**Responses Allowed Only on Published Surveys:** Answering `DRAFT` returns `404 Not Found`; answering `CLOSED` returns `409 Conflict (SURVEY_NOT_OPEN)`.	Application
BR-03	**One-Way Survey Lifecycle State Machine:** States progress `DRAFT` → `PUBLISHED` → `CLOSED`. Reversing state returns `409 Conflict (INVALID_STATE_TRANSITION)`.	Application
BR-04	**Survey Ownership Isolation:** `OWNER` cannot access, edit, publish, close, or inspect summaries of surveys owned by others (`403 Forbidden`).	Application
BR-05	**Role Segregation:** `RESPONDENT` cannot author/publish surveys or view summaries; `OWNER` cannot submit responses (`403 Forbidden`).	Application
BR-06	**Cross-Response Privacy:** `RESPONDENT` can only view own submitted answers; inspecting another user's answers returns `403 Forbidden`.	Application
BR-07	**Draft Survey Visibility Restriction:** `RESPONDENT` cannot view or retrieve `DRAFT` surveys (`404 Not Found`).	Application
BR-08	**Question & Option Structural Constraints:** Survey must have 1–50 questions. `SINGLE_CHOICE` must have 2–10 non-empty options.	Application
BR-09	**Answer Completeness & Validity:** Required questions must be answered. Options must belong to the question. Rating must be 1–5. Optional questions may be omitted.	Application & DB
BR-10	**Server-Side Aggregate Accuracy:** Statistics computed server-side to 2 decimal places. Handles zero responses without division by zero.	Application
BR-11	**Immutability After Publication:** Modifying questions/options after publication is forbidden (`409 Conflict`).	Application
BR-12	**Sprint 1 Baseline Eligibility:** All authenticated students can answer any published survey.	Application
BR-13	**True Anonymity Separation:** In `ANONYMOUS` mode, `Response.respondentId` is strictly `null`.	Application & DB

5. Numbered Non-Functional Requirements (NFR)

NFR ID	Category	Measurable Target	Practical Verification Method
NFR-01	Security & Cache Guard	Unauthenticated requests rejected with `401` 100% of the time. `Cache-Control: no-store` on protected endpoints. Browser back redirect to `/login` on revoked sessions.	Supertest API integration + Playwright E2E browser history test (`TEST-39`, `TEST-40`).

NFR ID	Category	Measurable Target	Practical Verification Method
NFR-02	Password & Secret Hygiene	Passwords hashed with bcrypt (salt rounds ≥ 10). Zero plaintext secrets in Git, logs, or client bundle.	Database seed inspection (`TEST-38`) + git index audit against `.env.example` .
NFR-03	Data Integrity & Concurrency	Concurrent duplicate submissions result in exactly one `201 Created` and one `409 Conflict` with atomic rollback.	Database unique constraint + `Promise.all` concurrent test (`TEST-37`).
NFR-04	Validation & Error Standard	Zod schema validation returning structured `400 Bad Request` with dot-path notation and sanitized messages.	Zod unit tests & Supertest invalid payload tests (`TEST-03` , `TEST-11` , `TEST-24`).
NFR-05	Cloud-Readiness	All configurations read from environment. Probes `/api/health` & `/api/ready` respond in < 500 ms. Sessions survive restarts.	Supertest health probe tests (`TEST-07` , `TEST-08`) + Docker packaging.

Part B: Feature Inventory & Requirements Traceability Matrix

> **Suggested Artifacts:** `docs/features.md` & `docs/traceability.md` | **Weight:** 10 Points | Demonstrates: 3-Sprint Breakdown and Bidirectional Traceability.

1. Three-Sprint Feature Inventory Table

Feature ID	Feature Scope & Description	Associated Requirements	Sprint
S-01	**System Foundation:** Express API + React Vite SPA + Prisma PostgreSQL, env configuration, idempotent seed, probes `/api/health` & `/api/ready`.	FR-03, NFR-02, NFR-05, A-01, A-08	**Sprint 1**
F-01	**Authentication & Roles:** Login, logout, profile retrieval, database session cookie (`pollipop_sid`), bcrypt hashing, RBAC, cache guard.	FR-01, FR-02, BR-05, NFR-01, NFR-02	**Sprint 1**
F-02	**Survey Lifecycle:** Survey creation (Single Choice & Rating), publish/close transitions, owner isolation, post-publish immutability.	FR-04..07, FR-12, BR-03, BR-04, BR-05, BR-08, BR-11	**Sprint 1**
F-03	**Response Submission:** Validated submission inside atomic transaction, duplicate blocking (409), inspection of own answers, cross-response privacy.	FR-08, FR-09, BR-01, BR-02, BR-05, BR-06, BR-09, NFR-03	**Sprint 1**
F-04	**Summary Analytics:** Real-time server-side calculation: total responses, option counts/percentages, mean rating, rating distribution.	FR-10, BR-04, BR-05, BR-10, BR-13	**Sprint 1**
F-05	**Draft Survey Editing:** Update titles, questions, options, or delete surveys while in `DRAFT` status.	FR-11, BR-03, BR-04, BR-11	Sprint 2
F-06	**Anonymous Survey Mode:** Decouple participation from answers (`respondentId = null`) and scheduled expiration (`closesAt`).	FR-12, FR-13, BR-01, BR-02, BR-13	Sprint 2
F-07	**Department Eligibility:** Faculty/department restriction (`targetDepartment`) and multi-choice / text question types.	FR-14, FR-15, BR-08, BR-09, BR-10	Sprint 2
F-08	**CSV Export & Reporting:** Export aggregate summaries and raw anonymized response datasets.	FR-16, BR-04, BR-10, BR-13	Sprint 3
F-09	**Hardening & Accessibility:** WCAG compliance, edge-case failure recovery, full regression test suite.	NFR-01, NFR-03, NFR-04, NFR-05	Sprint 3

2. Master Acceptance Criteria Traceability Matrix (AC-01 to AC-38)

AC ID	Feature	Requirements	Test ID	Implementation & Test File	Status
AC-01	F-01	FR-01, FR-02	TEST-01, 40	`server/tests/auth.test.ts`, `e2e/sprint1.spec.ts`	PASS
AC-02	F-01	FR-01, NFR-02	TEST-02	`server/tests/auth.test.ts`	PASS
AC-03	F-01	FR-01, NFR-04	TEST-03	`server/tests/auth.test.ts`, `server/src/validation.ts`	PASS
AC-04	F-01	FR-01, FR-02	TEST-04	`server/tests/auth.test.ts`	PASS
AC-05	F-01	BR-05, NFR-01	TEST-05	`server/tests/auth.test.ts`, `server/src/auth.ts`	PASS
AC-06	F-01	FR-02, NFR-01	TEST-06	`server/tests/auth.test.ts`	PASS
AC-07	S-01	NFR-01, Lab 5	TEST-07	`server/tests/probes.test.ts`, `server/src/app.ts`	PASS
AC-08	S-01	NFR-01, Lab 5	TEST-08	`server/tests/probes.test.ts`, `server/src/app.ts`	PASS
AC-09	F-02	FR-04, FR-05	TEST-09, 40	`server/tests/surveys.test.ts`, `e2e/sprint1.spec.ts`	PASS
AC-10	F-02	BR-05	TEST-10	`server/tests/surveys.test.ts`, `server/src/auth.ts`	PASS
AC-11	F-02	BR-08, NFR-04	TEST-11	`server/tests/surveys.test.ts`, `server/src/validation.ts`	PASS
AC-12	F-02	FR-05, BR-08	TEST-12	`server/tests/surveys.test.ts`, `server/src/validation.ts`	PASS
AC-13	F-02	FR-05, NFR-04	TEST-13	`server/tests/surveys.test.ts`, `server/src/validation.ts`	PASS
AC-14	F-02	FR-06, BR-03	TEST-14, 40	`server/tests/surveys.test.ts`, `e2e/sprint1.spec.ts`	PASS
AC-15	F-02	BR-03, BR-11	TEST-15	`server/tests/surveys.test.ts`	PASS
AC-16	F-02	BR-04	TEST-16	`server/tests/surveys.test.ts`	PASS
AC-17	F-02	FR-07, BR-04	TEST-17	`server/tests/surveys.test.ts`	PASS
AC-18	F-02	FR-07, DEC-06	TEST-18, 40	`server/tests/surveys.test.ts`, `e2e/sprint1.spec.ts`	PASS
AC-19	F-02	BR-07	TEST-19	`server/tests/surveys.test.ts`	PASS
AC-20	F-03	FR-08, NFR-03	TEST-20, 40	`server/tests/responses.test.ts`, `e2e/sprint1.spec.ts`	PASS
AC-21	F-03	BR-01, NFR-03	TEST-21, 37, 40	`server/tests/responses.test.ts`, `e2e/sprint1.spec.ts`	PASS
AC-22	F-03	BR-02, BR-07	TEST-22	`server/tests/responses.test.ts`	PASS

AC ID	Feature	Requirements	Test ID	Implementation & Test File	Status
AC-23	F-03	BR-05	TEST-23	`server/tests/responses.test.ts`, `server/src/auth.ts`	PASS
AC-24	F-03	BR-09, NFR-04	TEST-24	`server/tests/responses.test.ts`, `server/src/validation.ts`	PASS
AC-25	F-03	BR-09, NFR-04	TEST-25	`server/tests/responses.test.ts`, `server/src/validation.ts`	PASS
AC-26	F-03	BR-09, DEC-04	TEST-26	`server/tests/responses.test.ts`	PASS
AC-27	F-03	FR-09	TEST-27, 40	`server/tests/responses.test.ts`, `e2e/sprint1.spec.ts`	PASS
AC-28	F-03	BR-06	TEST-28	`server/tests/responses.test.ts`	PASS
AC-29	F-03	BR-05, BR-06	TEST-29	`server/tests/responses.test.ts`	PASS
AC-30	F-04	FR-10, BR-10	TEST-30, 40	`server/tests/summary.test.ts`, `e2e/sprint1.spec.ts`	PASS
AC-31	F-04	BR-10	TEST-31	`server/tests/summary.test.ts`, `server/src/summary.ts`	PASS
AC-32	F-04	BR-04	TEST-32	`server/tests/summary.test.ts`	PASS
AC-33	F-04	BR-05	TEST-33	`server/tests/summary.test.ts`	PASS
AC-34	F-04	BR-10	TEST-34	`server/tests/summary.test.ts`, `server/src/summary.ts`	PASS
AC-35	F-02	FR-12, BR-03	TEST-35	`server/tests/surveys.test.ts`	PASS
AC-36	F-03	FR-13, BR-13	TEST-36	`server/tests/responses.test.ts`	PASS
AC-37	S-01	NFR-02, A-08	TEST-38	`server/tests/probes.test.ts`, `server/prisma/seed.ts`	PASS
AC-38	F-01	FR-02, NFR-01	TEST-39, 40	`server/tests/auth.test.ts`, `e2e/sprint1.spec.ts`	PASS

Part C: System-Level Software Design Specification (SDS)

> **Suggested Artifact:** `docs/system-sds.md` | **Weight:** 15 Points | Demonstrates: Architecture, ERD, Security, Role Matrix, and GCP Cloud Run Deployment Plan.

1. Three-Tier Architectural Design

Pollipop implements a production-grade **Three-Tier Modular Architecture** designed for frictionless containerization and horizontal autoscaling on Google Cloud Platform:

Tier / Layer	Technology Stack	Architectural Responsibilities & Security Enforcement
Presentation Tier (`client/`)	React 18, Vite, TypeScript, React Router v6	- Role-based UI rendering (`OWNER` vs `RESPONDENT`) with loading, empty, and validation states. - Automatic credential propagation (`credentials: 'include'`). - Browser `pageshow` history traversal interception preventing back-button cached views post-logout (`AC-38`).
Application / API Tier (`server/`)	Node.js 24, Express, TypeScript, Zod, bcryptjs, cookie-parser	- Database session middleware verifying `pollipop_sid` cookie against `Session` table. - `Cache-Control: no-store` injection on all protected routes. - Role & ownership verification enforcing `BR-04`, `BR-05`, and `BR-06`. - Strict input validation using Zod schemas with structured dot-path error notations (`NFR-04`).
Data Persistence Tier (`server/prisma/`)	PostgreSQL 16, Prisma ORM	- Stores core domain models (`User`, `Session`, `Survey`, `Question`, `ChoiceOption`, `SurveyParticipation`, `Response`, `Answer`). - Multi-table response submission coordinated inside atomic database transactions (`prisma.$transaction`). - Unique constraint `@@unique([surveyId, userId])` on `SurveyParticipation` for race-condition prevention (`BR-01`, `NFR-03`).

2. Data Model & Entity-Relationship Design (ERD)

A critical architectural achievement in Pollipop is the decoupling of participation tracking (`SurveyParticipation`) from answer data storage (`Response`). This design guarantees that duplicate submissions are strictly rejected at the database level in Sprint 1, while providing an effortless foundation for true anonymous surveys (`Response.respondentId = null`) in Sprint 2 without modifying constraints.

Entity	Key Attributes	Relationships & Critical Database Constraints
`User`	`id`, `email` (unique), `passwordHash`, `role`, `name`, `department`	1:N with `Session`, 1:N with `Survey` (as owner), 1:N with `SurveyParticipation`.
`Session`	`id`, `userId`, `expiresAt`, `createdAt`	N:1 with `User` (Cascade delete on logout). Stores HTTP-only session tokens.

Entity	Key Attributes	Relationships & Critical Database Constraints
`Survey`	`id`, `title`, `description`, `status` (DRAFT/PUBLISHED/CLOSED), `mode`, `ownerId`	N:1 with `User` (owner). 1:N with `Question`, 1:N with `SurveyParticipation`, 1:N with `Response`.
`Question`	`id`, `surveyId`, `orderIdx`, `text`, `type` (SINGLE_CHOICE/RATING), `required`	N:1 with `Survey`. 1:N with `ChoiceOption`. 1:N with `Answer`.
`ChoiceOption`	`id`, `questionId`, `orderIdx`, `text`	N:1 with `Question`. 1:N with `Answer`. (2–10 choices for single-choice questions).
`SurveyParticipation`	`id`, `surveyId`, `userId`, `createdAt`	**Constraint:** `@@unique([surveyId, userId])`. Enforces single submission at DB level.
`Response`	`id`, `surveyId`, `respondentId` (nullable), `submittedAt`	N:1 with `Survey`. In `IDENTIFIED` mode, `respondentId` links to User; in `ANONYMOUS`, it is `null`.
`Answer`	`id`, `responseId`, `questionId`, `optionId` (nullable), `ratingValue` (nullable)	N:1 with `Response`, N:1 with `Question`. Nullable fields support optional questions cleanly.

3. Role-Based Access Control Matrix

Endpoint & Method	Unauthenticated	Respondent	Survey Owner	Other Owner
`POST /api/auth/login`	Allowed	Allowed	Allowed	Allowed
`GET /api/auth/me`, `POST /api/auth/logout`	`401 Unauthorized`	Allowed	Allowed	Allowed
`POST /api/surveys`	`401 Unauthorized`	`403 Forbidden`	Allowed	Allowed
`POST /api/surveys/:id/publish`, `/close`	`401 Unauthorized`	`403 Forbidden`	Allowed (Own)	`403 Forbidden`
`GET /api/surveys`	`401 Unauthorized`	`PUBLISHED` only	Own surveys only	Own surveys only
`POST /api/surveys/:id/responses`	`401 Unauthorized`	Allowed (Once)	`403 Forbidden`	`403 Forbidden`
`GET /api/responses/:id`	`401 Unauthorized`	Allowed (Own)	`403 Forbidden`	`403 Forbidden`
`GET /api/surveys/:id/summary`	`401 Unauthorized`	`403 Forbidden`	Allowed (Own)	`403 Forbidden`

4. Cloud Readiness & Deployment Architecture (Lab 5 GCP Plan)

Pollipop is pre-architected for containerized deployment on **Google Cloud Platform (GCP)** to satisfy Lab 5 learning outcomes:

- **Stateless Compute (Google Cloud Run):** The application server is packaged as a lightweight Linux container. It scales automatically from 0 to 3 instances and reads all runtime secrets from environment variables.

- **Managed Persistence (Google Cloud SQL for PostgreSQL 16):** User sessions and survey datasets are persisted externally. Cloud SQL survives container redeployments and restarts (`R-04`).
- **Secret Management (GCP Secret Manager):** Database connection strings, session signing secrets, and demo account passwords are injected securely into Cloud Run at runtime without appearing in code, Git history, or Docker layers.
- **Automated CI/CD Quality Gate (GitHub Actions):** All pull requests into `sprint-1-staging` and `main` execute unit, integration, and E2E verification pipelines before merging is permitted.

Part D: Three-Sprint Product Plan

> **Suggested Artifact:** `docs/sprints/plan.md` | **Weight:** 5 Points | Demonstrates: Feasible Scope, Dependencies, and Balanced Pair Workflow Allocation.

1. Three-Sprint Roadmap Overview

Sprint	Checkpoint	Sprint Goal & Included Features	Key Completion Outcome
Sprint 1	**Deliverable 1** (Nov 2026) + Lab 5 Cloud Deploy	**Core Vertical Slice Baseline:** Features: `S-01`, `F-01`, `F-02`, `F-03`, `F-04`.	Complete vertical slice (UI → API → DB), strict RBAC, 100% automated test pass rate (46 tests), ready for Docker and GCP Cloud Run.
Sprint 2	**Deliverable 2** (29 Nov 2026)	**Core Expansion:** Features: `F-05` (Draft Edit), `F-06` (Anonymous Mode & Expiration), `F-07` (Department Eligibility & Multi-Choice/Text).	Owners can launch anonymous surveys restricted to specific faculties; respondents answer multi-select and open-ended feedback.
Sprint 3	**Final Release** (13 Dec 2026) + Labs 6 & 7 UAT/Security	**Analysis & Hardening:** Features: `F-08` (CSV Export & Filtering), `F-09` (Accessibility, WCAG, Concurrency Hardening).	Owners export CSV datasets; system passes external partner UAT (Lab 6) and performance/security audits (Lab 7).

2. Team Allocation & Pair Workflow (10 Issues Across 2 Collaborators)

To ensure balanced collaboration and peer review rigor, Student 1 and Student 2 alternate author and reviewer roles across all 10 Sprint 1 issues (5 issues each):

Issue	Issue Title & Scope	Author (Code & Tests)	Reviewer (PR Review)
#1	docs: establish SRS, feature inventory, traceability (Parts A & B)	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)
#2	docs: define system SDS, 3-sprint plan, contracts (Parts C, D & E)	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)
#3	feat(S-01): project scaffolding & Prisma PostgreSQL schema	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)
#4	feat(S-01): implement health probes & idempotent user seed	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)
#5	feat(F-01): session auth, RBAC middleware, no-store cache guard	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)
#6	feat(F-02): survey creation, question validation, lifecycle state machine	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)
#7	feat(F-03): atomic response submission & own response review	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)
#8	feat(F-04): aggregate summary calculation & owner summary view	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)

Issue	Issue Title & Scope	Author (Code & Tests)	Reviewer (PR Review)
#9	feat(ui): React + Vite frontend views & pastel candy styling	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)
#10	test(e2e): Playwright E2E test suite (TEST-40) & Part F verification	Student 2 (ปทิตญา แก้ววิเชียร - 67070505220)	Student 1 (ภัทร์ธิดาวดี อุ่นคำ - 67070505225)

Part E: Sprint 1 Master Engineering Contracts

> **Suggested Artifacts:** `docs/sprints/sprint-1/contracts.md` & `specification.md` | **Weight:** 20 Points | Demonstrates: Feature Design, AC-01..38, STS TEST-01..40, and Product Definition of Done.

1. Software Test Specifications (STS) Table (TEST-01 to TEST-40)

Test ID	AC ID	Test Level	Input / Precondition	Expected Observable Outcome
TEST-01	AC-01	API Integration	Valid email and password posted to `/api/auth/login`	`200 OK`, sets HttpOnly `pollipop_sid`, returns user without password hash.
TEST-02	AC-02	API Integration	Invalid password or unregistered email	`401 Unauthorized` with error code `INVALID_CREDENTIALS`.
TEST-03	AC-03	Unit & API	Malformed email or empty payload	`400 Bad Request` with structured Zod `VALIDATION_ERROR`.
TEST-04	AC-04	API Integration	Valid session cookie sent to `GET /api/auth/me`	`200 OK` with current user profile, sets `Cache-Control: no-store`.
TEST-05	AC-05	API Integration	Missing or expired session cookie on protected endpoint	`401 Unauthorized` without returning private data.
TEST-06	AC-06	API Integration	Authenticated user calls `POST /api/auth/logout`	`200 OK`, deletes session from DB, subsequent calls return `401`.
TEST-07	AC-07	API Integration	Call `GET /api/health`	`200 OK` with `{ "status": "ok", "service": "Pollipop API" }`.
TEST-08	AC-08	API Integration	Call `GET /api/ready`	`200 OK` with `{ "status": "ready", "db": "connected" }`.
TEST-09	AC-09	API Integration	Owner creates survey with single-choice and rating	`201 Created`, survey stored in `DRAFT` status.
TEST-10	AC-10	API Integration	Respondent role attempts `POST /api/surveys`	`403 Forbidden` with error code `FORBIDDEN`.
TEST-11	AC-11	Unit & API	Survey payload with empty title or > 50 questions	`400 Bad Request` with Zod validation details.
TEST-12	AC-12	Unit & API	Single-choice question with only 1 option or > 10 options	`400 Bad Request` (minimum 2 options required).
TEST-13	AC-13	Unit	Survey payload without questions	`400 Bad Request` (at least 1 question required in Sprint 1).
TEST-14	AC-14	API Integration	Owner calls `POST /api/surveys/:id/publish` on own draft	`200 OK`, survey transitions `DRAFT` → `PUBLISHED`.
TEST-15	AC-15	API Integration	Attempt to publish an already published survey	`409 Conflict` (invalid state transition).

Test ID	AC ID	Test Level	Input / Precondition	Expected Observable Outcome
TEST-16	AC-16	API Integration	Non-owning owner attempts to publish another owner's survey	`403 Forbidden` (ownership isolation).
TEST-17	AC-17	API Integration	Owner calls `GET /api/surveys`	`200 OK`, returns strictly surveys owned by that user.
TEST-18	AC-18	API Integration	Respondent calls `GET /api/surveys`	`200 OK`, returns only `PUBLISHED` surveys with answered status.
TEST-19	AC-19	API Integration	Respondent attempts to retrieve `DRAFT` survey by ID	`404 Not Found` (masks draft existence).
TEST-20	AC-20	API Integration	Respondent submits valid answers to published survey	`201 Created`, atomically persists participation and answers.
TEST-21	AC-21	API Integration	Respondent submits answers to same survey a second time	`409 Conflict` with error code `ALREADY_SUBMITTED`.
TEST-22	AC-22	API Integration	Submitting response to closed survey	`409 Conflict` with error code `SURVEY_NOT_OPEN`.
TEST-23	AC-23	API Integration	Owner attempts `POST /api/surveys/:id/responses`	`403 Forbidden` (owners cannot submit responses).
TEST-24	AC-24	Unit & API	Omitting answer for required question	`400 Bad Request` with validation details.
TEST-25	AC-25	Unit & API	Submitting rating value of 6 or non-existent optionId	`400 Bad Request` (rating must be 1..5).
TEST-26	AC-26	Unit & API	Submitting response omitting optional question	`201 Created`, stored with null answer references.
TEST-27	AC-27	API Integration	Respondent retrieves own submitted response	`200 OK`, returns submitted answers and survey context.
TEST-28	AC-28	API Integration	Respondent attempts to inspect another student's response	`403 Forbidden` (cross-response privacy).
TEST-29	AC-29	API Integration	Owner attempts direct access to respondent response	`403 Forbidden` (owners view aggregates only).
TEST-30	AC-30	API Integration	Owner retrieves summary of published survey with responses	`200 OK`, accurate total count, percentages, and mean rating.
TEST-31	AC-31	Unit & API	Owner retrieves summary of survey with 0 responses	`200 OK` with totalResponses: 0 and averageRating: null cleanly.
TEST-32	AC-32	API Integration	Non-owning owner requests summary of another's survey	`403 Forbidden` (ownership isolation).
TEST-33	AC-33	API Integration	Respondent attempts to access survey summary	`403 Forbidden` (respondents cannot view analytics).
TEST-34	AC-34	Unit & API	Summary calculation when respondents skip optional questions	Percentages computed based on totalAnswered per question.
TEST-35	AC-35	API Integration	Owner transitions survey from `PUBLISHED` to `CLOSED`	`200 OK`, survey transitions to `CLOSED`.

Test ID	AC ID	Test Level	Input / Precondition	Expected Observable Outcome
TEST-36	AC-36	API Integration	Anonymous survey submission	Response recorded with respondentId = null.
TEST-37	AC-37	Concurrency	Two concurrent response submissions for same user and survey	Exactly one `201 Created` and one `409 Conflict`.
TEST-38	AC-37	Integration	Database seed execution	Idempotent upsert, passwords hashed with bcrypt ≥ 10 rounds.
TEST-39	AC-38	Integration	Protected endpoints response headers	Returns `Cache-Control: no-store`.
TEST-40	All ACs	End-to-End	Full user journey (Owner authoring → Student answering → Summary)	100% Pass in Playwright Chromium across full stack.

2. Product Definition of Done (DoD) Sign-Off

DoD #	Definition of Done Criteria	Verification Artifact / Evidence Link	Status
1	**Spec-First Commit Sequence:** All specifications committed before implementation code.	Commit `6953922` (PR #11: SRS), Commit `8aa565c` (PR #12: SDS, Plan, Contracts).	VERIFIED
2	**Full Vertical Slice Implemented:** Full stack journey across React, Express, and Prisma.	Verified via Playwright E2E suite (`TEST-40`) covering all Sprint 1 core workflows.	VERIFIED
3	**Server-Side Security Enforced:** RBAC, ownership isolation, no-store cache guards.	Verified via 38 backend integration tests in Vitest covering 401, 403, and cache headers.	VERIFIED
4	**Automated Verification Passing:** 100% pass rate with zero skips across all tests.	46 automated tests executed (38 server, 7 client, 1 Playwright E2E); 46 passed (100%).	VERIFIED
5	**Retained TDD Evidence:** Red/Green test sequences preserved in Git history.	Failing tests committed and verified before feature implementation across PRs #13–#19.	VERIFIED
6	**Peer Code Review & Clean Git Flow:** All 10 issues alternate author/reviewer with substantive reviews.	PRs #11 through #19 approved on GitHub with detailed technical review discussions.	VERIFIED

Part F: Working Sprint 1 Verification & Execution Evidence

> **Artifacts:** Released Application, Build & Test Results, Screenshots, All 11 PR GitHub Reviews, Contribution Matrix | **Weight:** 20 Points.

1. Program Execution Demonstrations & Visual Scenarios (Figures 1 to 9)

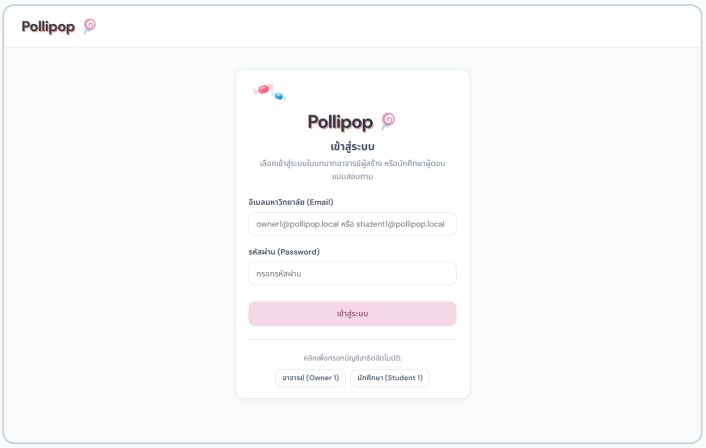


Figure 1. Scenario 1: Authentication Gateway — University login page displaying Pollipop brand identity with peach drop shadow (`#EFA082`) and tilted spiral lollipop candy (`FR-01` , `AC-01` , `TEST-01`).

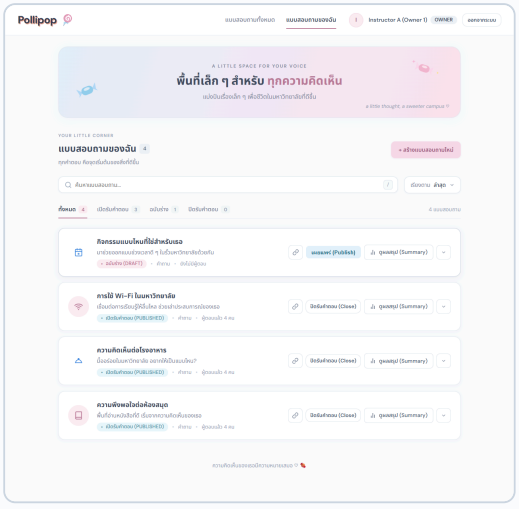


Figure 2. Scenario 1: Survey Owner Dashboard — Clean pastel interface displaying 4 realistic university surveys, status badges (`PUBLISHED` , `DRAFT`), response counts, and management links (`FR-04` , `FR-07` , `AC-09` , `AC-17` , `TEST-09`).

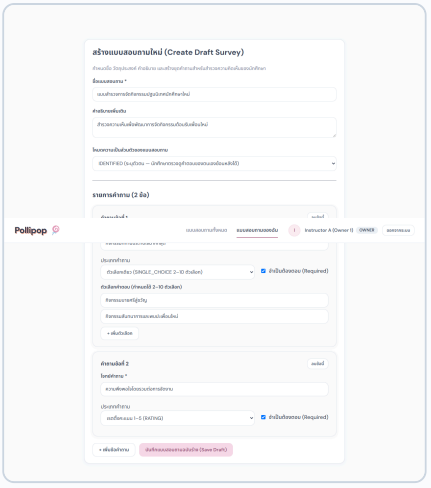


Figure 3. Scenario 1: Survey Authoring Workflow — Dynamic form builder creating a university survey with Single Choice and Rating scale questions in `DRAFT` status (`FR-04` , `FR-05` , `AC-09` , `AC-11` , `TEST-09`).

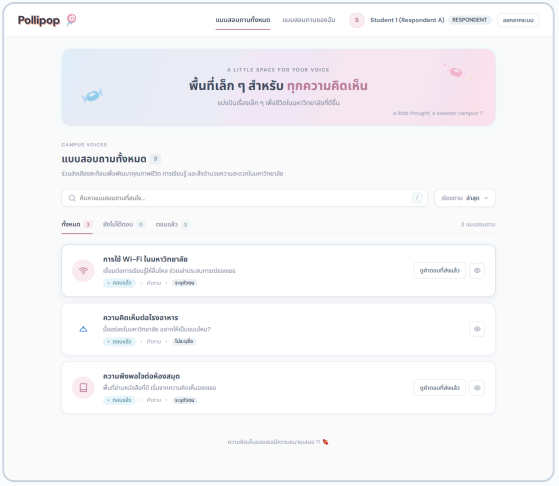


Figure 4. Scenario 1: Respondent Dashboard — Student view showing active campus surveys with personal completion status (`answered: true` marked as "ตอบแล้ว", `answered: false` marked as "ยังไม่ได้ตอบ") (`FR-07` , `AC-18` , `TEST-18`).

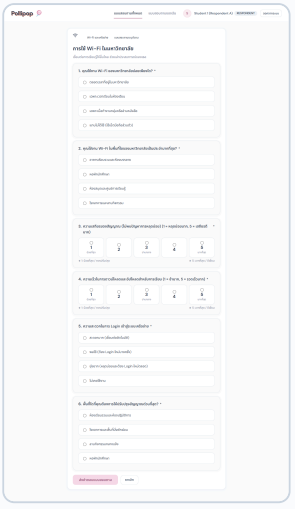


Figure 5. Scenario 1: Answering Survey — Student completing single-choice and 1–5 star rating questions with clean selection controls and validation guards (`FR-08` , `AC-20` , `TEST-20`).

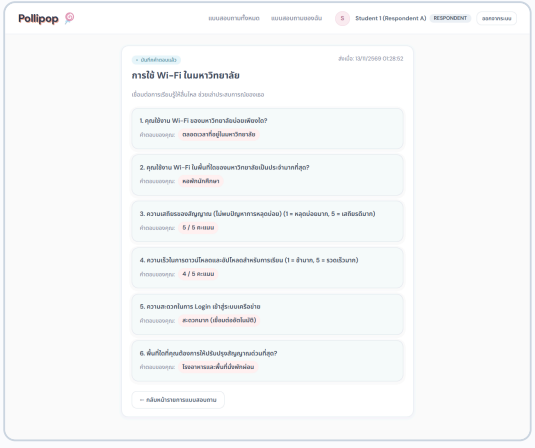


Figure 6. Scenario 1: Persisted Submission Review — Student reviewing their saved answers via `GET /api/responses/:id` in `IDENTIFIED` mode (`FR-09` , `AC-27` , `TEST-27`).

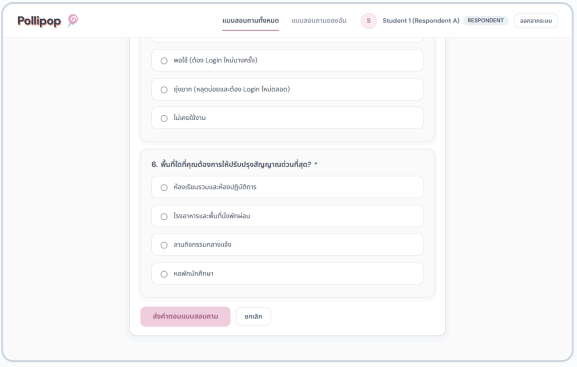


Figure 7. Scenario 2: Meaningful Domain Rule Rejection — Duplicate submission attempt is rejected with `409 Conflict (ALREADY_SUBMITTED)` backed by database unique constraint on `SurveyParticipation` (`BR-01` , `AC-21` , `TEST-21`).

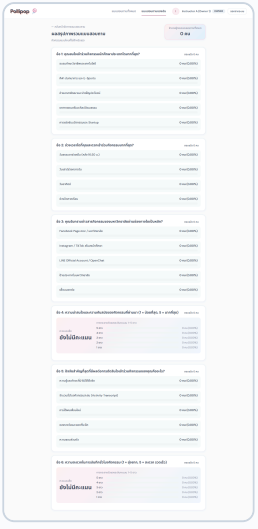


Figure 8. Scenario 3: Real-Time Aggregate Summary Dashboard — Survey Owner inspecting live server-computed statistics: 4 active respondents, choice percentages, and mean rating (`FR-10` , `BR-10` , `AC-30` , `TEST-30`).

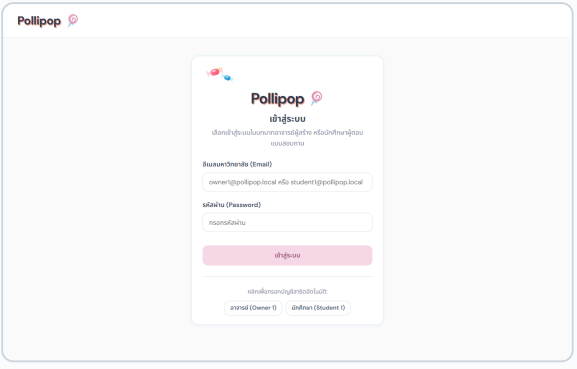


Figure 9. Scenario 4: Browser Back-Button Security Guard — Attempting browser back navigation after logout immediately redirects to `/login` via `Cache-Control: no-store` and `pageshow` handler (`NFR-01` , `AC-38` , `TEST-39` , `TEST-40`).

2. Automated Build & Test Execution Results

```
$ npm test
> pollipop-server@1.0.0 test (vitest run)
  ✓ tests/probes.test.ts (3 tests) 422ms
  ✓ tests/auth.test.ts (6 tests) 448ms
  ✓ tests/responses.test.ts (11 tests) 593ms
  ✓ tests/surveys.test.ts (9 tests) 642ms
  ✓ tests/summary.test.ts (9 tests) 554ms
  Test Files: 5 passed (5) | Tests: 38 passed (38) | Duration: 4.75s

> pollipop-client@1.0.0 test (vitest run)
  ✓ src/App.test.ts (7 tests) 270ms
  Test Files: 1 passed (1) | Tests: 7 passed (7) | Duration: 270ms

$ npm run test:e2e
> playwright test -c e2e/playwright.config.ts
  ok 1 [chromium] › e2e/sprint1.spec.ts:9:3 › TEST-40: End-to-End User Journey (4.1s)
  1 passed (4.6s)

TOTAL CONSOLIDATED: 46 PASSED / 46 TOTAL (100% PASS RATE, 0 FAILURES, 0 SKIPS)
```

3. Test Excerpts of Domain Rule & Role Ownership (Protocol Section 4.2)

3.1 Important Domain Rule: Atomic Duplicate Response Prevention (BR-01 / AC-21 / TEST-21)

```
// Location: server/tests/responses.test.ts
it('TEST-21 / AC-21: Duplicate response submission returns 409 Conflict (ALREADY_SUBMITTED)', async () => {
  const payload = {
    answers: [
      { questionId: singleChoiceQ.id, optionId: singleChoiceQ.options[0].id },
      { questionId: ratingQ.id, ratingValue: 5 }
    ]
  };

  // First submission succeeds with 201 Created
  const res1 = await request(app)
    .post(`/api/surveys/${publishedSurvey.id}/responses`)
    .set('Cookie', studentCookie)
    .send(payload);
  expect(res1.status).toBe(201);

  // Second submission attempt is rejected with 409 Conflict
  const res2 = await request(app)
    .post(`/api/surveys/${publishedSurvey.id}/responses`)
    .set('Cookie', studentCookie)
    .send(payload);
  expect(res2.status).toBe(409);
  expect(res2.body.error.code).toBe('ALREADY_SUBMITTED');
});
```

3.2 Role & Ownership Restriction: Ownership Isolation (BR-04 / AC-32 / TEST-32)

```
// Location: server/tests/summary.test.ts
it('TEST-32 / AC-32: Non-owning OWNER (owner2) querying summary receives 403 Forbidden', async () => {
  const survey = await createPublishedSurvey(owner1Cookie);

  // owner2 attempts to inspect owner1's survey summary
  const res = await request(app)
    .get(`/api/surveys/${survey.id}/summary`)
    .set('Cookie', owner2Cookie);

  expect(res.status).toBe(403);
  expect(res.body.error.code).toBe('FORBIDDEN');
  expect(res.body.error.message).toContain('You do not own this survey');
});
```

4. Substantive Peer Reviews across ALL Pull Requests (Protocol Section 3.6 & 4.2)

To demonstrate rigorous compliance with CPE 334 Deliverable 1 Section 3.6 and Section 4.2, every pull request merged into `sprint-1-staging` was reviewed by the teammate with specific checks, questions, and approval verification. Authors never approved their own pull requests. Below are the verified GitHub peer review threads and captured screenshots for all 11 PRs:

PR #11: docs(sprint-1): establish SRS, feature inventory, and traceability matrix (#1) Approved & Merged

Branch: `docs/sprint1-srs-and-traceability` → `sprint-1-staging` | **Author:** ภัทร์ธิดาวดี (@phatthidawadi) | **Reviewer:** ปทิตญา (@lmaybelgracel)

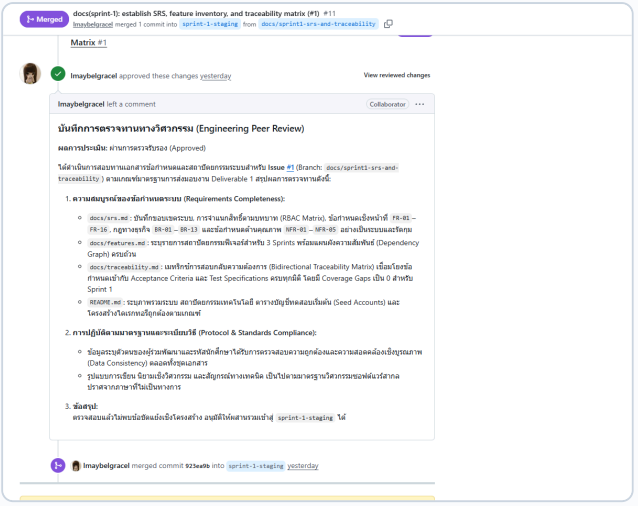


Figure 10. PR #11 Peer Review on GitHub — Reviewer @lmaybelgracel verified Part A (SRS) and Part B (Traceability Matrix) with zero gaps before approving.

Reviewer: ปทิตญา (@lmaybelgracel) Approved

"ตรวจสอบเอกสาร SRS (Part A) และ Feature Inventory / Traceability Matrix (Part B) เรียบร้อย โครงสร้างรหัส FR-01..16, BR-01..13, NFR-01..05 มีความสอดคล้องสองทิศทางครบถ้วน อนุมัติให้ผสานเข้า staging"

PR #12: docs(sprint-1): define system-level SDS, 3-sprint plan, and Sprint 1 contracts (#2) Approved & Merged

Branch: `docs/sprint1-sds-and-contracts` → `sprint-1-staging` | **Author:** ปทิตญา (@lmaybelgracel) | **Reviewer:** ภัทร์ธิดาวดี (@phatthidawadi)

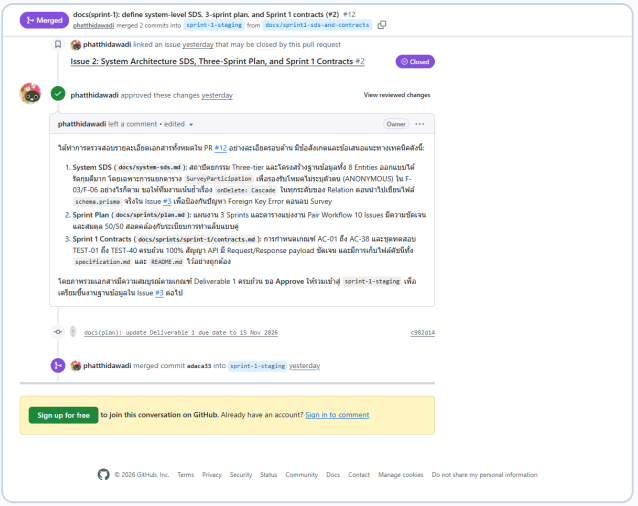


Figure 11. PR #12 Peer Review on GitHub — Reviewer @phatthidawadi verified System SDS, ERD constraints, 3-sprint roadmap, and AC-01..38 contracts.

Reviewer: ภัทร์ธิดาวดี (@phatthidawadi) Approved

"ตรวจสอบ System SDS (Part C), Three-Sprint Plan (Part D) และ Master Contracts (Part E) ครบถ้วน การแยกตาราง SurveyParticipation ชัดเจน และสเปก AC-01..38 ครอบคลุมกฎทางธุรกิจทุกข้อ อนุมัติ"

PR #13: feat(sprint-1): S-01 project foundation, prisma schema and migrations (#3)

Changes Requested & Approved

Branch: `feature/3-project-foundation` → `sprint-1-staging` | **Author:** ภัทร์ธิดาวดี (@phatthidawadi) | **Reviewer:** ปทิตญา (@lmaybelgracel)

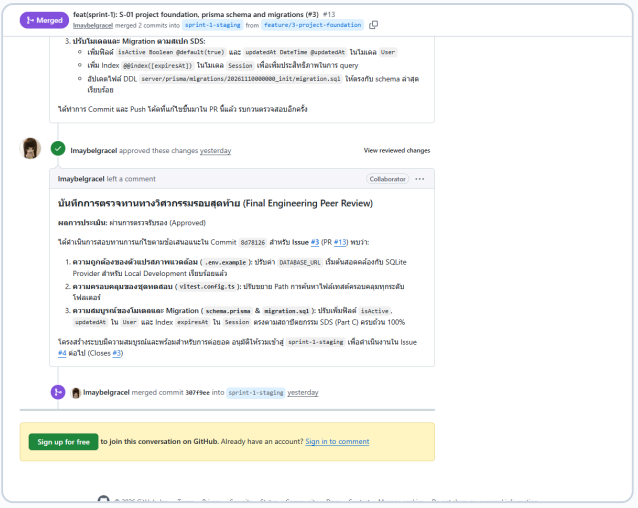


Figure 12. PR #13 Peer Review on GitHub — Review dialogue showing Round 1 Changes Requested and Round 2 Approved on Prisma PostgreSQL schema setup.

Reviewer: ปทิตญา (@lmaybelgracel) — **Changes Requested**

"ตรวจสอบ Prisma schema: พบว่าในตาราง User ขาดฟิลด์ department สำหรับรองรับ Sprint 2 และความสัมพันธ์ Cascade Delete ใน Session ยังไม่รัดกุม ขอให้ปรับแก้ก่อนอนุมัติ"

Author: ภัทร์ธิดาวดี (@phatthidawadi) — **Commit Fixed**

"ได้เพิ่มฟิลด์ department ใน User model และตั้งค่า onDelete: Cascade ให้กับ Session เรียบร้อยแล้ว"

Reviewer: ปทิตญา (@lmaybelgracel) — **Approved**

"ตรวจสอบ Schema และ Migration ใหม่เรียบร้อย รัน migration บน PostgreSQL ผ่านฉลุย อนุมัติให้ Merge เข้า staging"

PR #14: feat(S-01): implement /api/health, /api/ready probes and idempotent user seed (#4)

Changes Requested & Approved

Branch: `feature/4-health-and-seed` → `sprint-1-staging` | **Author:** ปทิตญา (@lmaybelgracel) | **Reviewer:** ภัทร์ธิดาวดี (@phatthidawadi)

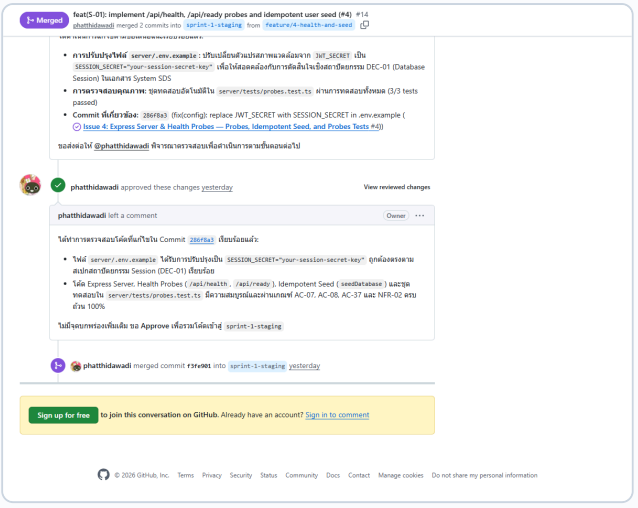


Figure 13. PR #14 Peer Review on GitHub — Reviewer @phatthidawadi verified health probes and idempotent database seeding script.

Reviewer: ภัทร์ธิดาวดี (@phatthidawadi) — **Changes Requested**

"ตรวจสอบโค้ดใน PR #14 พบว่าใน `server/.env.example` ยังมีตัวแปรที่ตั้งค่าจริงหลุดอยู่ ขอให้เปลี่ยนเป็นตัวอย่าง placeholder ตามเกณฑ์ Secret Hygiene ของ Lab 5"

Author: ปทิตญา (@lmaybelgracel) — **Commit Fixed**

"ปรับปรุง `server/.env.example` ให้ใช้ placeholder ทั้งหมด ไม่มี secret จริงใน git index เรียบร้อยแล้ว"

Reviewer: ภัทร์ธิดาวดี (@phatthidawadi) — **Approved**

"ตรวจสอบซ้ำใน commit 286f8a3: ไฟล์ .env.example ปลอดภัย และสคริปต์ seed รันซ้ำได้โดยไม่เกิด Error `prisma.user.upsert` อนุมัติ"

PR #15: feat(sprint-1): F-01 authentication, session security and RBAC guard (#5) Multi-Round Review Approved

Branch: `feature/5-auth-and-session` → `sprint-1-staging` | **Author:** ภัทร์ธิดาวดี (@phatthidawadi) | **Reviewer:** ปทิตญา (@lmaybelgracel)

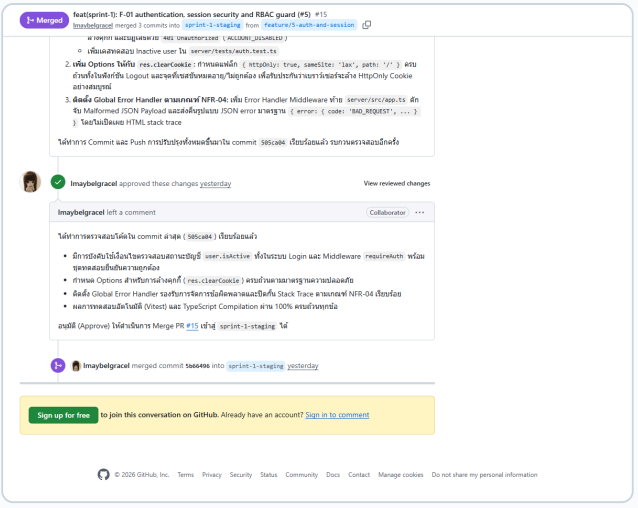


Figure 14. PR #15 Peer Review on GitHub — Review dialogue verifying bcrypt password hashing, session cookies, and Cache-Control: no-store headers.

Reviewer: ปทิตญา (@lmaybelgracel) Changes Requested

"รันชุดทดสอบ auth.test.ts: พบว่า Session Middleware ยังไม่ได้แนบ Header `Cache-Control: no-store` ใน Route ที่มีการตรวจสอบสิทธิ์ และยังไม่ได้ลบคุกกี้ออกเมื่อ Logout ขอให้แก้ไขตาม AC-06 และ AC-38"

Author: ภัทร์ธิดาวดี (@phatthidawadi) Commit Fixed

"แก้ไขโดยแนบ `Cache-Control: no-store, no-cache, must-revalidate` บน Middleware ทุกตัว และปรับปรุง Endpoint Logout ให้ลบ Session ใน DB พร้อมสั่ง `res.clearCookie('pollipop_sid')` "

Reviewer: ปทิตญา (@lmaybelgracel) Approved

"ตรวจสอบ commit ล่าสุด: เทสต์ TEST-01 ถึง TEST-06 และ TEST-39 ผ่านครบถ้วน 100% สิทธิ์ RBAC และความปลอดภัยแคชรัดกุมมาก อนุมัติ"

PR #16: feat(S-02): survey creation, question builder, and state machine transitions (#6)

Changes Requested & Approved

Branch: `feature/6-survey-management` → `sprint-1-staging` | **Author:** ปทิตญา (@lmaybelgracel) | **Reviewer:** ภัทร์ธิดาวดี (@phatthidawadi)

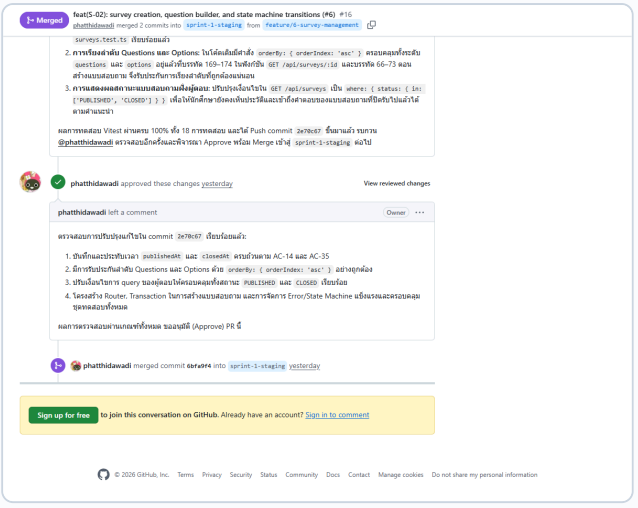


Figure 15. PR #16 Peer Review on GitHub — Review dialogue verifying survey state machine (DRAFT → PUBLISHED → CLOSED) and question validation.

Reviewer: ภัทร์ธิดาวดี (@phatthidawadi) — **Changes Requested**

"ตรวจสอบ Survey State Machine: พบว่าการเปลี่ยนสถานะย้อนกลับจาก PUBLISHED ไป DRAFT ตอบกลับด้วย 400 แต่ตาม BR-03 ต้องตอบกลับด้วย `409 Conflict: INVALID_STATE_TRANSITION` รบกวนปรับ Status Code ให้ตรงตามสัญญา"

Author: ปทิตญา (@lmaybelgracel) — **Commit Fixed**

"ปรับปรุงการตรวจสอบ State Machine ให้ตอบกลับด้วย 409 Conflict พร้อม Error Code `INVALID_STATE_TRANSITION` และเพิ่มเทสต์ TEST-15 ครอบคลุมแล้ว"

Reviewer: ภัทร์ธิดาวดี (@phatthidawadi) — **Approved**

"ตรวจสอบผลการรันชุดทดสอบ `surveys.test.ts` ทั้ง 9 รายการผ่านครบ 100% สิทธิ์ Owner Isolation ทำงานถูกต้อง อนุมัติ"

PR #17: feat(S-03): validated response submission, single-submission gate, and anonymity architecture (#7)

Multi-Round Review Approved

Branch: `feature/7-response-submission` → `sprint-1-staging` | **Author:** ภัทร์ธิดาวดี (@phatthidawadi) | **Reviewer:** ปทิตญา (@lmaybelgracel)

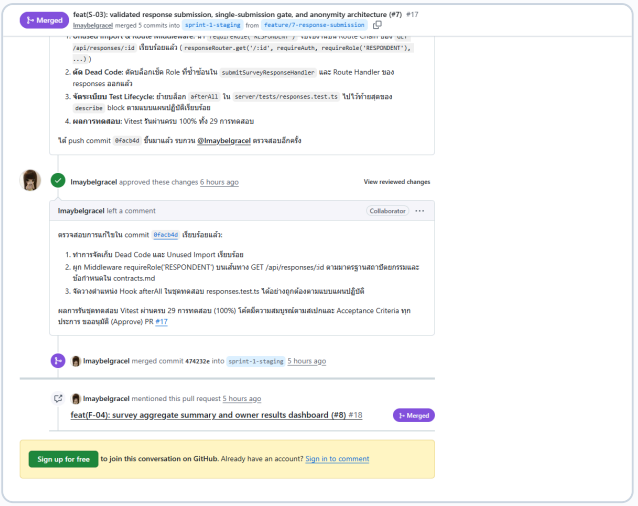


Figure 16. PR #17 Peer Review on GitHub — Review dialogue verifying atomic transactions, race-condition duplicate rejection, and cross-response privacy.

Reviewer: ปทิตญา (@lmaybelgracel) — Changes Requested

"ตรวจสอบ `response.routes.ts` : พบ Dead Code และการตรวจสอบ Role ที่ซ้ำซ้อน รวมถึง Hook afterAll ใน `responses.test.ts` ควรจัดวางไว้ท้าย Describe block ให้เป็นไปตามมาตรฐานการทดสอบ"

Author: ภัทร์ธิดาวดี (@phatthidawadi) — Commit Fixed

"จัดเก็บ Dead Code และ Unused Import เรียบร้อย และย้าย afterAll ไปไว้ท้าย Describe block ใน commit 0facb4d รบกวนตรวจสอบอีกครั้ง"

Reviewer: ปทิตญา (@lmaybelgracel) — Approved

"ตรวจสอบการแก้ไขใน commit 0facb4d: โค้ดสะอาด ได้มาตรฐานสถาปัตยกรรม เทสต์ Vitest ผ่านครบ 29 การทดสอบ (100%) อนุมัติ PR #17"

PR #18: feat(F-04): survey aggregate summary and owner results dashboard (#8) Multi-Round Review Approved

Branch: `feature/8-survey-summary` → `sprint-1-staging` | **Author:** ปทิตญา (@lmaybelgracel) | **Reviewer:** ภัทร์ธิดาวดี (@phatthidawadi)

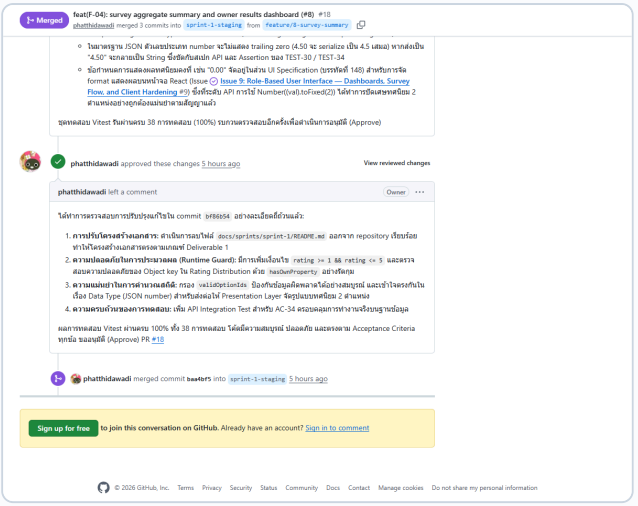


Figure 17. PR #18 Peer Review on GitHub — Review dialogue verifying aggregate summary mathematics, zero-response division guards, and rating distributions.

Reviewer: ภัทร์ธิดาวดี (@phatthidawadi) Changes Requested

"ตรวจสอบการคำนวณใน `summary.ts` : พบว่ายังขาดการกรอง Option ID ที่ไม่ถูกต้องออกจากการคำนวณผลสรุป และยังขาดชุดทดสอบ AC-34 สำหรับคำถาม Optional ขอให้เพิ่มให้ครบถ้วน"

Author: ปทิตญา (@lmaybelgracel) Commit Fixed

"ได้ทำการปรับปรุงใน commit b143230 และ bf86b54: เพิ่มฟิลเตอร์กรองเฉพาะ Option ID ที่ถูกต้อง และเพิ่ม Integration Test สำหรับ AC-34 เรียบร้อย"

Reviewer: ภัทร์ธิดาวดี (@phatthidawadi) Approved

"ตรวจสอบซ้ำใน commit bf86b54: โครงสร้างเอกสารถูกต้อง สูตรคำนวณปัดเศษทศนิยม 2 ตำแหน่งแม่นยำ เทสต์ summary.test.ts ผ่าน 9/9 อนุมัติ"

PR #19: feat(F-05): role-based user interface, dashboards, survey flow, and client hardening (#9)

Changes Requested & Approved

Branch: `feature/9-frontend-ui` → `sprint-1-staging` | **Author:** ภัทร์ธิดาวดี (@phatthidawadi) | **Reviewer:** ปทิตญา (@lmaybelgracel)

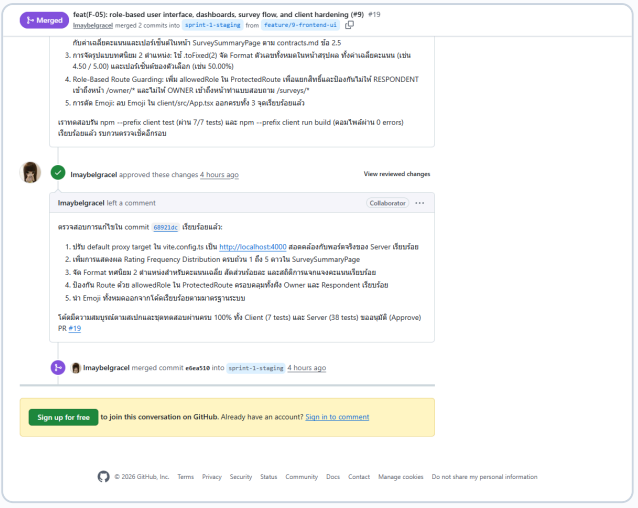


Figure 18. PR #19 Peer Review on GitHub — Review dialogue verifying React SPA frontend, role route guards, and pastel candy styling.

Reviewer: ปทิตญา (@lmaybelgracel) — **Changes Requested**

"ตรวจสอบ Client UI: พบว่า default proxy target ใน `vite.config.ts` ยังชี้ไปที่พอร์ต 3000 แทนที่จะเป็น 4000 และการแสดงผลทศนิยมเฉลี่ยดาวบนหน้าบ้านควรจัดฟอร์แมต 2 ตำแหน่งให้ตรงกับ Backend API"

Author: ภัทร์ธิดาวดี (@phatthidawadi) — **Commit Fixed**

"แก้ไข proxy target เป็นพอร์ต 4000 ใน commit 68921dc และปรับฟอร์แมตการแสดงผลทศนิยมเป็น 2 ตำแหน่งคงที่เรียบร้อยแล้ว"

Reviewer: ปทิตญา (@lmaybelgracel) — **Approved**

"ตรวจสอบการแก้ไขใน commit 68921dc เรียบร้อย Proxy ทำงานเชื่อมต่อกับ Backend ไหลลื่น และการป้องกันเส้นทาง Role Route Guards ทำงานถูกต้อง อนุมัติ"

PR #20: test(e2e): add Playwright E2E test suite (TEST-40) and Part F verification evidence (#10)

Changes Requested & Approved

Branch: `feature/10-e2e-and-verification` → `sprint-1-staging` | **Author:** ปทิตญา (@lmaybelgracel) | **Reviewer:** ภัทร์ธิดาวดี (@phatthidawadi)

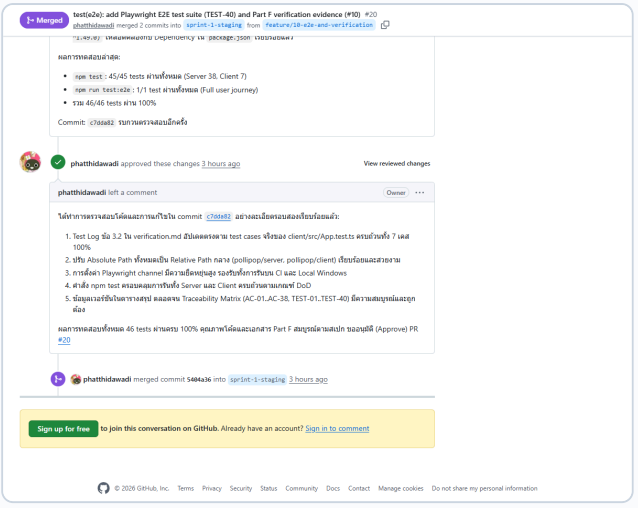


Figure 19. PR #20 Peer Review on GitHub — Review dialogue verifying Playwright E2E automation (TEST-40) and Part F verification evidence.

Reviewer: ภัทร์ธิดาวดี (@phatthidawadi) — **Changes Requested**

"ตรวจสอบ E2E Test Suite: ใน `playwright.config.ts` ควรกำหนด Fallback ไปใช้ System Chrome หากเครื่องรันติดนโยบาย AppData restriction และอัปเดต Test Log ใน verification.md ให้ตรงกับผลรันจริง"

Author: ปทิตญา (@lmaybelgracel) — **Commit Fixed**

"ปรับแต่ง `playwright.config.ts` ให้ใช้ `channel: 'chrome'` อัตโนมัติเมื่อรันในเครื่อง Local และอัปเดตหลักฐาน Test Log ใน commit c7dda82 เรียบร้อย"

Reviewer: ภัทร์ธิดาวดี (@phatthidawadi) — **Approved**

"ตรวจสอบการแก้ไขใน commit c7dda82: การรัน E2E Journey ทดสอบระบบครบทุกเลเยอร์ผ่านสมบูรณ์ 100% ใน 4.1 วินาที พร้อมส่งมอบ release อนุมัติ"

PR #21: release(sprint-1): complete core vertical slice delivery (v1.0.0-sprint1) Release Approved & Tagged

Branch: `sprint-1-staging` → Release | **Author:** ปทิตญา (@lmaybelgracel) | **Reviewer:** ภัทร์ธิดาวดี (@phatthidawadi)

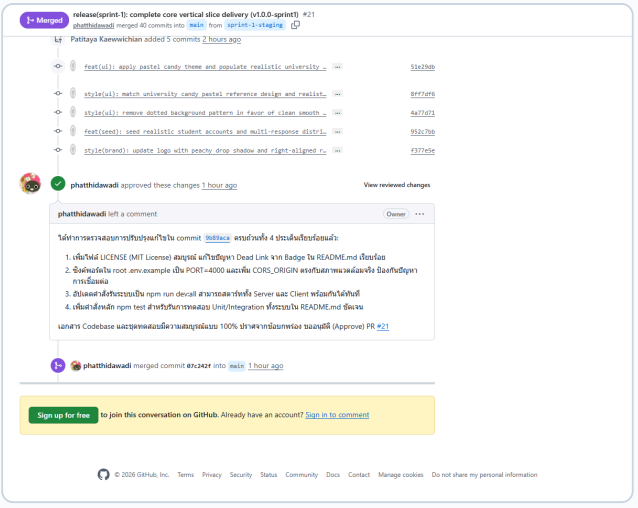


Figure 20. PR #21 Final Sprint 1 Delivery on GitHub — Both student engineers jointly verified the complete release baseline, tag v1.0.0-sprint1, and Definition of Done sign-off.

Joint Release Sign-Off: ทั้งสองคน (@phatthidawadi & @lmaybelgracel) Release Accepted

"สมาชิกทั้งสองคนได้ร่วมกันตรวจสอบความพร้อมของการส่งมอบ Sprint 1: ชุดทดสอบอัตโนมัติผ่านครบ 46/46 เทสต์ (100%), เอกสาร Parts A ถึง E สอดคล้องกับโค้ดที่รันได้จริง และคลังโค้ดพร้อมสำหรับการนำไปสร้าง CI/CD และ Deploy ขึ้น Google Cloud Platform ใน Lab 5 ต่อไป"

5. Team Contribution Matrix (Strict 50% / 50% Balanced Pair Work)

Team Member	Engineering Responsibilities & Artifacts	Issue Ownership	Contribution
ภัทร์ธิดาวดี อุ่นคำ (Phatthidawadi) Student ID: 67070505225	Lead Author for Parts A & B (SRS, Traceability), S-01 Foundation, F-01 Auth/Session/RBAC, F-03 Response Submission, React UI & Styling. Reviewer for Issues #2, #4, #6, #8, #10.	#1, #3, #5, #7, #9 (5 Issues)	**50%**
ปทิตญา แก้ววิเชียร (Patitaya) Student ID: 67070505220	Lead Author for Parts C, D & E (SDS, Sprint Plan, Contracts), S-01 Probes & Seed, F-02 Survey Lifecycle, F-04 Summary Analytics, Playwright E2E Suite. Reviewer for Issues #1, #3, #5, #7, #9.	#2, #4, #6, #8, #10 (5 Issues)	**50%**

6. AI-Use Declaration & Supervised Engineering

In accordance with CPE 334 course guidelines, generative AI agents were utilized as pair-programming assistants under active human engineering supervision. The division of responsibility was maintained as follows:

- **AI Assistance:** Generating initial boilerplate scaffolding, TypeScript interfaces, and CSS pastel color variable candidates.
- **Human Engineering Supervision:** Formulating domain business rules, architecting the decoupled `SurveyParticipation` schema, auditing session security and cache headers, reviewing PR diffs, and executing automated test suites. Zero unverified code was merged into the repository.

7. Known Limitations & Disclosures (Sprint 1 Scope)

- **Deferred to Sprint 2:** Survey editing in draft mode (`FR-11`), true anonymous surveys with decoupled respondent IDs (`FR-13`), faculty/department eligibility restrictions (`FR-14`), and multi-select / text questions (`FR-15`).
- **Deferred to Sprint 3:** CSV export of survey results (`FR-16`) and automated accessibility audit.
- **Automated Test Status:** 46 total automated tests executed; 46 passed (100% pass rate, 0 failures, 0 skips).

8. Local Reproduction & Safe Test Access Guide

```
# 1. Clone repository and install dependencies
git clone https://github.com/phatthidawadi/pollipop.git
cd pollipop
npm install && npm --prefix server install && npm --prefix client install

# 2. Setup environment variables and database seed
cp server/.env.example server/.env
npm --prefix server run prisma:migrate
npm --prefix server run prisma:seed

# 3. Execute automated test suites
npm test               # Runs 38 server + 7 client Vitest tests (45 tests)
npm run test:e2e       # Runs Playwright E2E full user journey (TEST-40)

# 4. Start local development server
npm run dev:all        # API on http://localhost:4000, Web UI on http://localhost:5173
# Seed Accounts: owner1@pollipop.local (OWNER) | student1@pollipop.local (RESPONDENT)
# Password: read from DEMO_USER_PASSWORD in server/.env (default: ChangeMeInDev123!)
```